KSJ RESEARCH

Research Note Vol.4

Selling Power to a Data Centre

Equity in a hypothetical 35 MW solar asset — KEPCO contract (CfD), direct PPA with a data centre, or merchant: which pays for its risk?

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The data centre's solar-hour saving is KRW 151.3 — below the seller's required 152.8–164.2, so the direct PPA fails before any network charge

Price evidence (official first) → cash flows and debt for three routes → break-evens (price, termination, network charges) → Monte Carlo → IC memo

Key metrics [*1,*2,*3,*4,*5]

Key metric	Value	p.
Hypothetical solar · CAPEX (backed out of LCOE, judgement)	35MW · KRW 61.1bn	1
Yr-1 output · share of data-centre use	49.1 GWh · 15%	1
Corridor: KEPCO contract ~ solar-hour saving	**147.7 ~ 151.3**	3
Saving · spring-autumn / summer / winter	126.9 / 167.0 / 172.6	5
Equity IRR · CfD / PPA (mid) / merchant	**12.2% / 11.6% / 3.7%**	4
Gearing · CfD / PPA / merchant	80% / 75% / 38%	4
Required PPA premium (1% term., 0–2pp spread, NPV)	**+5.1 ~ +16.5**	5
Buyer's room for network charges (solar-hour saving)	**−12.9 ~ −1.4**	5
Public example of charges (3rd-party PPA, 2021)	**~KRW 40**	3
Monte Carlo · PPA equity IRR P5 / P50	5.5% / 11.3%	7
Monte Carlo · merchant loss probability (IRR < 0)	5%	7

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *4 SFOC, 'Stranded RE100' (Dec 2021, citing KEPCO's third-party PPA charge simulator) *5 KSJ Research
Note: KSJ Research. Asset and offtaker are hypothetical. Prices KRW/kWh, amounts KRW 100mn. Source grades and provisional inputs in A2–A4

11.6%

Equity IRR · direct PPA (corridor mid KRW 149.5/kWh, no termination)

Same plant, different buyer: equity IRR Merchant 3.7% → Direct PPA 11.6% → KEPCO CfD 12.2%

- KEPCO: the 2026 auction cap KRW 147.7/kWh as a proxy for the 2027 contract market → gearing 80%, IRR 12.2%
- Merchant: SMP KRW 113.2/kWh (12-month mainland average), no RECs for post-2027 entrants → 3.7%
- IRR levels depend on CAPEX (backed out of LCOE, judgement) — at 2022 actual cost KEPCO IRR is 23.7%; the price gap between routes barely moves

+5~16

Required PPA premium (1% termination, PPA discount +0–2pp, per kWh)

Price to accept a PPA instead of KEPCO: KRW 152.8/kWh–KRW 164.2/kWh

- Rule (NPV): PPA price at which expected equity NPV equals KEPCO's — PPA flows discounted at the hurdle plus a 0–2pp risk spread; SMP after termination
- Judgement that moves it: PPA gearing 65% → +9, 80% → +3 (spread 0)
- Termination rate and spread have no public basis, so grids are shown; a deal sets them from the offtaker's credit

−13 ~ −1

Buyer's room for network and other charges (solar-hour saving 151.3, per kWh)

Solar-hour saving 151.3 − required PPA = room −12.9 ~ −1.4 — the buyer loses even with zero network charges

- Buyer ceiling = post-reform (16 Apr 2026) industrial (B) HV-A option II energy rates in solar hours, output-weighted — demand charges are not avoided. On the 181.9 average tariff the room would be 18~29
- Seasons split: spring/autumn 126.9 · summer 167.0 · winter 172.6 — spring/autumn below the KEPCO price 147.7
- To close, the buyer must pay extra (RE100 value) = required + charges − saving → 11–23 with the 9.73 KPX settlement alone, 41–53 at the 40 example

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PPA equity IRR P50 11.3% · P5 5.5% — below KEPCO 99% of the time; merchant loses money 5% of the time

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Measured on solar-hour rates the price itself blocks the deal — confirm the buyer's RE100 willingness to pay first, then charges and termination protection

App. **Appendix** A1–A5

A1 IC memo · A2–A3 Inputs and source grades · A4 Data notes, method · A5 Glossary

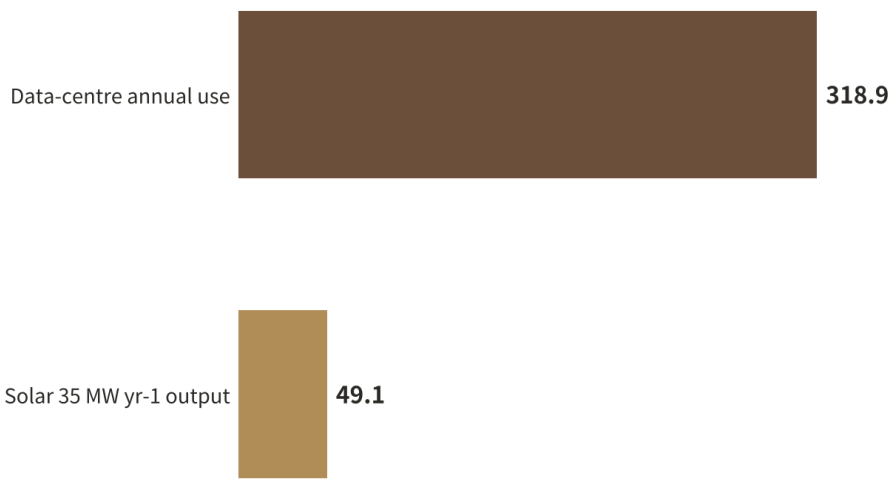
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Starting point: a 40 MW data centre's power bill is KRW 58.0bn a year — 35 MW of solar sized within its load covers 15%

Vol.2 data centre (lender's view) → Vol.4 the plant that sells it power (equity investor's view)

Data centre 318.9 GWh a year vs solar 49.1 GWh — 15% supplied directly [*1,*2]

Unit: GWh



Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KSJ Research
Note: Vol.2 Korean case (IT 40 MW). 35 MW of solar = peak output within the data centre's 36.4 MW average load, so all output is consumed; nights and gaps come from KEPCO

Power bill KRW 58.0bn a year — the cost Vol.2 passed to tenants [*1,*2,*3,*4]

Step	Value
IT capacity × load × PUE	40MW × 70% × 1.3
× 8,760 h = annual use	318.9 GWh
× KEPCO industrial average tariff (2025)	KRW 181.9/kWh
= Annual power bill	**KRW 58.0bn**
Solar MW × capacity factor × 8,760 h	35MW × 16%
= Yr-1 output (0.5% annual degradation)	**49.1 GWh**

Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 NREL Jordan & Kurtz (2012), PV degradation review *4 KSJ Research
Note: For the annual bill = average tariff incl. demand charges. A PPA saves less — the solar-hour energy rate, KRW 151.3/kWh (p3, p5)

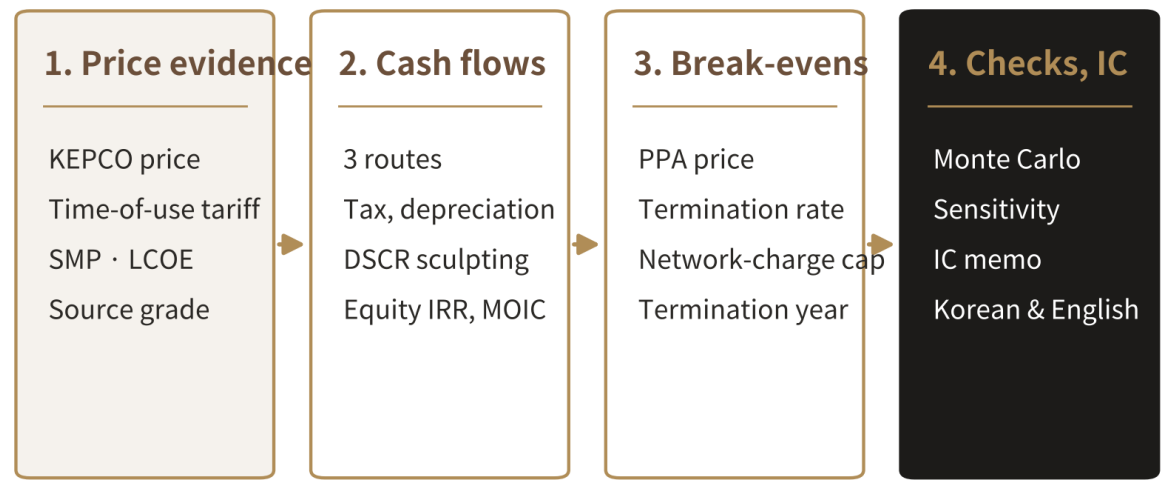
Question: if the plant sells to the data centre instead of KEPCO, does the extra price pay for the risk? [*1,*2]

An equity investor's three routes

- ① KEPCO contract (CfD): standard for new solar from 2027; 20-yr fixed price, KEPCO as counterparty
- ② Direct PPA: negotiated price; counterparty is Vol.2's data-centre SPV
- ③ Merchant (SMP): full price risk; no RECs for new entrants
- Vol.2 took the lender's view, Vol.3 the ECM view — this note takes the equity investor's

Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KSJ Research

Structure: prices → cash flows → break-evens → checks and IC memo [*1]



Source: *1 KSJ Research

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Rules: from 2027 new solar defaults to a 20-yr KEPCO contract — a PPA is the alternative, merchant sales get no RECs

RPS repeal · contract market · distributed-energy cap · AI data-centre act → three routes

Timeline: from 2027 a KEPCO long-term contract is the default, a direct PPA the alternative [*1,*2,*3]

Date	Change	Grade
2026-02-03	Distributed-energy zones: firms on local grids to be allowed renewable PPAs; 35 MW cap under review	Press report / secondary report
2026-05-07	AI Data Centre Act passed — the requested LNG direct-PPA carve-out was dropped	Press report / secondary report
2026-05	Distributed resources capped at 40 MW in principle; a data-centre carve-out bill is pending (ministry reported opposed)	Press report / secondary report
2026-07-16	2026 round-1 solar auction: cap KRW 147.686/kWh (about 5% lower), ~1,000 MW	Official source
2026-09-15	Renewable Energy Act amended — RPS ends; a single long-term contract market with KEPCO	Press report / secondary report
2027-01-01	Effective — RECs restricted for new entrants; REC spot market until end-2029	Press report / secondary report

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Financial News (15 Sep 2026, citing the ministry): Renewable Energy Act reform *3 Herald K (9 May 2026) · Hankyung (3 Feb 2026): distributed energy, AI data-centre act
Note: Grades: official = government release checked · press/outside report = media citing a government release (original not yet matched) — dates and content as reported

Three routes: price, counterparty and debt terms move together [*1,*2,*3]

Item	KEPCO contract (CfD)	Direct PPA with data centre	Merchant (SMP)
Price	Auction price fixed 20 yrs (cap 147.7)	Negotiated between KEPCO price and buyer's saving	SMP (no RECs)
Counterparty	KEPCO	Data-centre SPV (Vol.2)	Market
Term	20 yrs	20 yrs (U.S. nuclear PPA precedents)	—
Target DSCR · max gearing	1.20 · 80%	1.30 · 75%	1.60 · 55%
Tenor	18 yrs	18 yrs	12 yrs
Equity risk	Capped upside · losing the auction	Offtaker credit · network charges · termination	All price risk

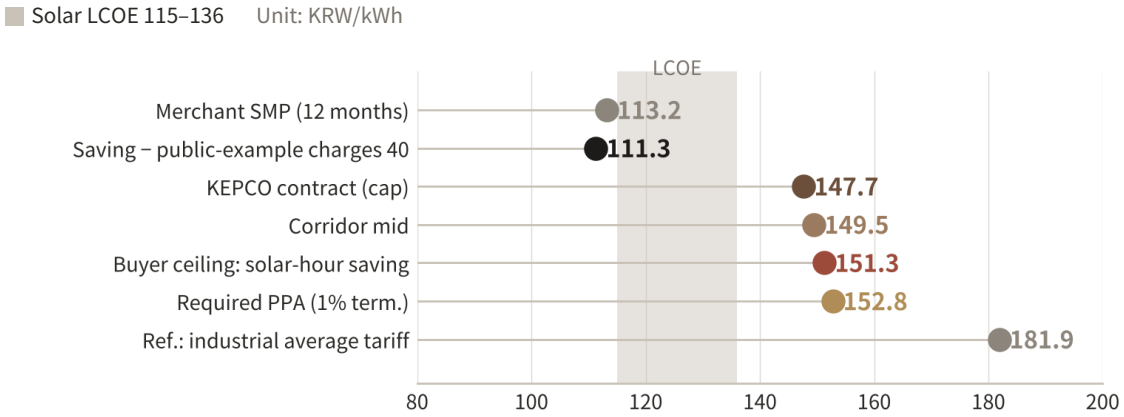
Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Financial News (15 Sep 2026, citing the ministry): Renewable Energy Act reform *3 KSJ Research
Note: Debt terms are author judgments (no public term sheets): weaker counterparty → higher DSCR, lower gearing. Rate = KTB 3-yr 3.84% (2026-10-20) + 150bp

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Price corridor: KEPCO 147.7 to the solar-hour saving 151.3 — 30.6 narrower than on the average tariff (181.9)

Floor = seller's alternative · ceiling = buyer's alternative (solar-hour rate) · cost = LCOE · merchant = SMP · U.S. check

Corridor 147.7–151.3 (width 3.6) — the required PPA price 152.8 already exceeds the buyer's ceiling, before any charges *1,*2,*3,*4,*5,*6,*7,*8



Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566, table 16-1 (2025 monthly solar output) *4 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *5 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *6 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *7 SFOC, 'Stranded RE100' (Dec 2021, citing KEPCO's third-party PPA charge simulator) *8 KSJ Research
Note: KEPCO price = 2026 round-1 cap (proxy for 2027). Buyer ceiling = post-reform industrial (B) HV-A option II energy rates in solar hours, weighted by 2025 monthly output (no demand charge; daily shape is a judgement). Average tariff = reference only. Public example = 2021 third-party PPA simulation. LCOE = 2024 Korean range · Provisional: Daily output shape exponent, Saturday time-band rule, Per-kWh add-ons (climate, fuel adjustment) pending official source (A3)

Price evidence: 7 items — official 4 · judgement 2 · outside report 1 *1,*2,*3,*4,*5,*6,*7

Item (KRW/kWh)	Value (range)	Grade	Published
KEPCO contract price (2026 round-1 cap)	147.7	Author judgement	2026-07-16
Buyer ceiling: solar-hour saving (HV-A option II)	151.3 (149.8~152.0)	Author judgement	2026-03-13
KEPCO industrial average tariff (2025, reference)	181.9	Official source	2026-02
Solar LCOE	125.5 (115~136)	Official source	2025
SMP base	113.2 (90~148)	Official source	2026-10
Direct-PPA ancillary settlement (Jul 2023–Jun 2024)	9.7	Official source	2023-12-19
Ancillary charges, public example (3rd-party PPA, 2021)	40.0	Press report / secondary report	2021-12

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *4 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *5 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *6 KPX Q&A (19 Dec 2023): direct-PPA ancillary settlement rate *7 SFOC, 'Stranded RE100' (Dec 2021, citing KEPCO's third-party PPA charge simulator)
Note: Saving = official tariff (13 Mar 2026) and 2025 monthly output (official) with a judgement daily shape (range = shape sensitivity) · Average tariff = 2025 industrial revenue ÷ volume (reference) · SMP = mean of monthly mainland SMP, Sep 2025–Aug 2026 (cross-checked with 10:00–16:00 hourly)

U.S. check: hyperscaler nuclear PPA at least $63/MWh — Korean solar contracts cost more *1,*2,*3

Item	USD/MWh	Grade
Talen–AWS nuclear PPA (17 yrs, 1,920 MW)	≥ 63	Official source
LevelTen N. America solar PPA avg (Q2 2026)	61.4	Press report / secondary report
Ref.: Korea KEPCO price (USD/KRW 1,332.1)	111	Author judgement
Ref.: Korea buyer's solar-hour saving	114	Author judgement

Source: *1 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *2 TaiyangNews (23 Jul 2026, citing LevelTen Q2 2026) *3 Bank of Korea ECOS KTB 3-yr, USD/KRW
Note: Talen: ~USD 18bn notional ÷ (1,920 MW × 8,760 h × 17 yrs) = average price at full, round-the-clock volume; the ramp to 2032 means the true price is higher (floor). Other nuclear PPAs (e.g. Meta–Constellation) are undisclosed

How to read it: the seller's floor is KEPCO's contract, the buyer's ceiling KEPCO's solar-hour rate *1

A PPA price lives between two outside options

- Seller: needs more than KEPCO (147.7) for extra risk → floor + risk premium
- Buyer: PPA + charges must beat the solar-hour rate (151.3); demand charges stay. With the 40 example, the PPA must be below 111.3
- Merchant SMP (113.2) sits below the KEPCO price — no RECs for post-2027 entrants
- U.S. 24/7 nuclear at $63+/MWh — a Korean PPA is an RE100 premium product, not cheap power

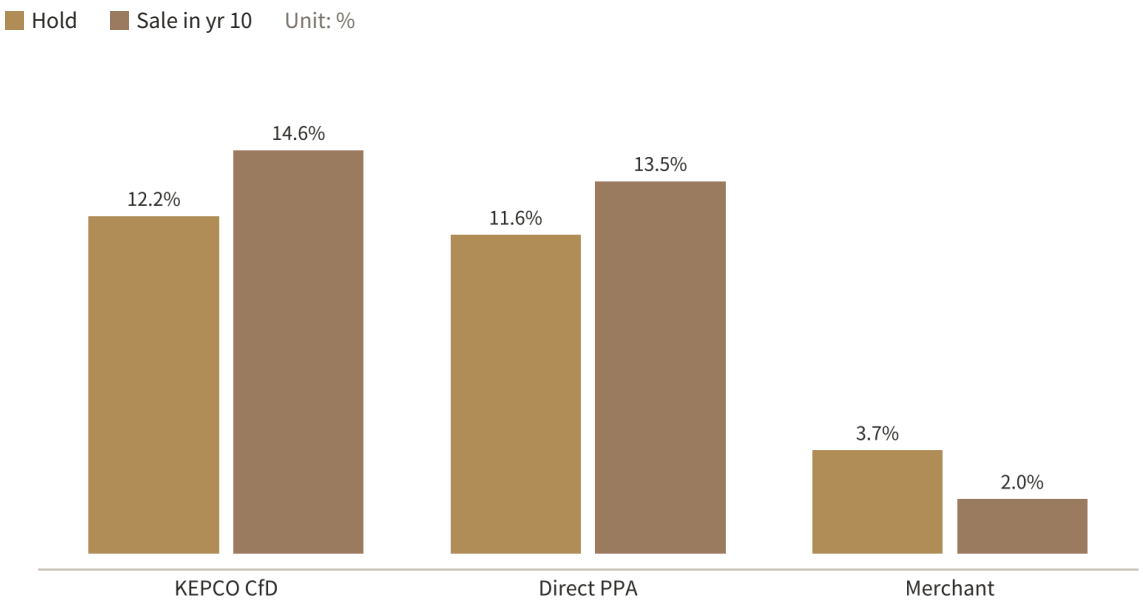
Source: *1 KSJ Research

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Equity IRR: PPA 11.6% · KEPCO 12.2% · merchant 3.7% — against the 8% hurdle: KEPCO CfD · Direct PPA clear, Merchant miss

Debt sizing by route → equity IRR and MOIC → exit → sensitivity

Equity IRR: hold (25 yrs) vs sale in year 10 — KEPCO CfD leads either way [*1]



Source: *1 KSJ Research
Note: PPA at corridor mid KRW 149.5/kWh, no termination. Sale price = later equity flows at the buyer's 7%

Debt sizing: the stronger the counterparty, the more debt and the less equity [*1,*2,*3]

Route	Debt	Gearing	Min DSCR	Equity	Eq. IRR	MOIC
KEPCO CfD	KRW 48.9bn	80%	1.33x	KRW 12.2bn	12.2%	3.96x
Direct PPA	KRW 45.8bn	75%	1.43x	KRW 15.3bn	**11.6%**	3.54x
Merchant	KRW 23.4bn	38%	1.60x	KRW 37.7bn	3.7%	1.67x

Source: *1 Bank of Korea ECOS KTB 3-yr, USD/KRW *2 Corporate Tax Act art. 55 · Local Tax Act (law.go.kr) *3 KSJ Research
Note: CAPEX KRW 61.1bn (KRW 1,746k/kW, backed out of LCOE/WACC/CF — judgement) · progressive corporate tax + 10% local · straight-line 20 yrs · debt = min(DSCR sculpting, max gearing)

Sensitivity: CAPEX and capacity factor move IRR more than SMP — contracted routes carry cost risk [*1,*2,*3]

Input (low–high)	KEPCO CfD	Direct PPA	Merchant
CAPEX (KRW k/kW) 1,396–1,892	23.7% ~ 8.8%	21.1% ~ 8.7%	7.3% ~ 2.7%
Capacity factor 14%–18%	6.6% ~ 17.9%	6.7% ~ 16.3%	2.0% ~ 5.5%
SMP base 90–148	11.9% ~ 12.7%	11.3% ~ 12.0%	0.9% ~ 8.2%
OPEX (% of CAPEX) 1%–2%	14.5% ~ 9.9%	13.5% ~ 9.6%	4.7% ~ 2.7%

Source: *1 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *2 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *3 KSJ Research
Note: One input at a time at its range ends (debt re-sized). CAPEX low = 2022 actual cost, high = matched to the 136 LCOE · CF range is an author sensitivity

How to read it: at the corridor mid (+1.8 over KEPCO) the PPA cannot beat KEPCO's leverage [*1]

Same plant, different capital structures

- KEPCO: price 147.7 vs LCOE 125.5, 80% debt → 12.2%
- PPA: 11.6% at the corridor mid KRW 149.5/kWh (no termination) — it needs KRW 152.3/kWh to match KEPCO (p5)
- Merchant: lenders give only 38% and SMP is low → 3.7%
- Against the 8% hurdle: KEPCO CfD · Direct PPA clear, Merchant miss — between contracts it is price vs termination risk

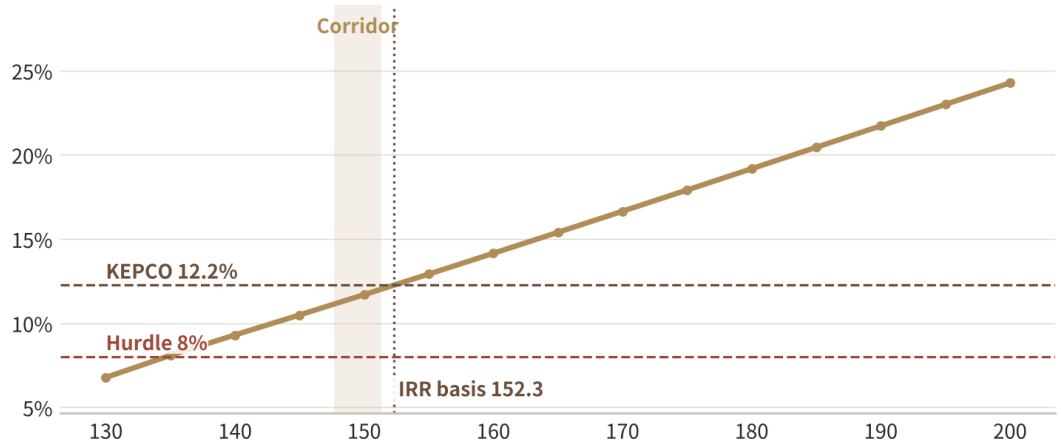
Source: *1 KSJ Research

PPA negotiation: the seller needs +5–16 over KEPCO, the buyer saves only +3.6 — room before charges is −12.9 ~ −1.4

IRR by price → required price by termination × spread (NPV) → buyer's seasonal saving → what it takes to close

Equity IRR by PPA price: equals KEPCO's IRR at KRW 152.3 — above the buyer's 151.3 ceiling *1,*2,*3

Unit: Equity IRR (no termination)



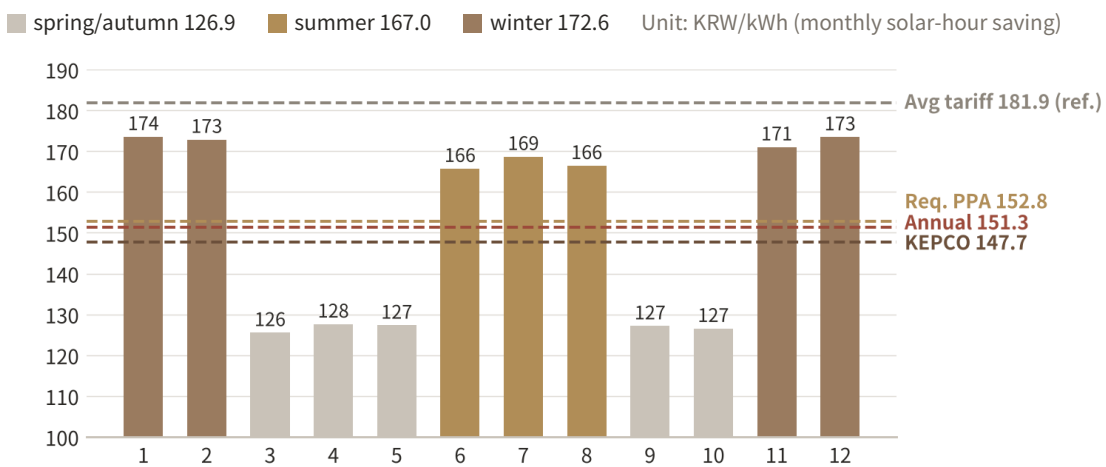
Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KSJ Research
Note: Debt re-sized at each price (PPA terms). All 'required prices' here are NPV-based (table) — 149.7 with no termination and no spread. The IRR basis (152.3) is higher because equity sizes differ (gearing 80% vs 75%)

Required premium: +5–16 at 1% termination — the risk spread matters as much as the termination rate *1,*2,*3,*4

Annual term.	Premium +0pp	Premium +1pp	Premium +2pp	Room for charges
0.0%	+2.0	+7.4	+12.8	−9.2 ~ 1.6
0.5%	+3.5	+9.1	+14.6	−11.0 ~ 0.1
1.0%	**+5.1**	**+10.8**	**+16.5**	**−12.9 ~ −1.4**
2.0%	+8.3	+14.3	+20.3	−16.7 ~ −4.7
3.0%	+11.6	+18.0	+24.3	−20.7 ~ −8.0
5.0%	+18.8	+25.9	+32.8	−29.2 ~ −15.2

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566, table 16-1 (2025 monthly solar output) *4 KSJ Research
Note: Required price (NPV) = PPA price at which expected equity NPV equals KEPCO's equity NPV at 8%. PPA flows discounted at the hurdle plus a spread (private-counterparty premium, judgement grid). Constant annual termination during the PPA, SMP afterwards, debt kept at no-termination sizing. Premium = required − KEPCO price; room = solar-hour saving 151.3 − required (negative = buyer loses even with zero charges)

Seasonal saving vs KEPCO price: spring/autumn (43% of output) at 126.9, below KEPCO's 147.7 *1,*2,*3,*4,*5



Source: *1 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *2 KEPCO Monthly Statistics no.566, table 16-1 (2025 monthly solar output) *3 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *4 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *5 KSJ Research
Note: Post-reform industrial (B) HV-A option II energy rates × daily output shape (judgement) × 2025 calendar (Sundays off-peak all day, 50% off 11:00–14:00 on spring/autumn weekends); seasonal and annual values weighted by 2025 monthly output. Required PPA at 1% termination, no spread. If the discount ends after 2030: 154.1 a year · Provisional: Daily output shape exponent, Saturday time-band rule, Sunday/holiday rate rule, Per-kWh add-ons (climate, fuel adjustment) pending official source (A3)

The negotiation: on solar-hour rates the price itself does not close — it turns on what the buyer pays for RE100 value *1,*2,*3

What the IC will ask

- A 3.6 corridor vs the seller's +5–16 → buyer room −12.9 ~ −1.4 (40 public example)
- To close, the buyer must pay 11–23 (KPX settlement only) to 41–53 (public example) per kWh for RE100 value — the 181.9 average tariff hides this
- Judgement that moves it: PPA gearing 65% → +9, 80% → +3 (1%, no spread)
- In a corporate survey, opaque network-charge setting is the No.2 reason to avoid PPAs (45.2%); KPX has commissioned a direct-PPA policy study (Newsis, 4 May 2026)

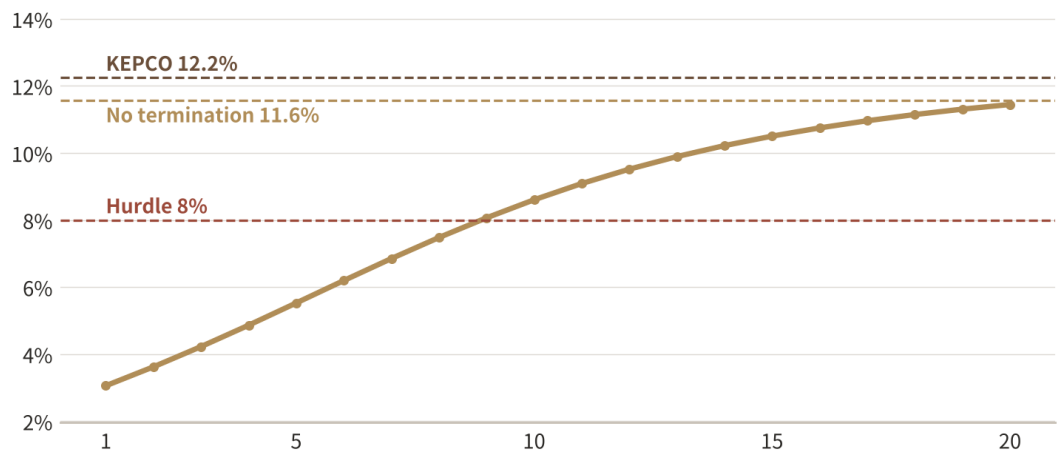
Source: *1 Newsis (4 May 2026): SFOC corporate survey, KPX direct-PPA study *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KSJ Research

Offtaker risk: at the corridor mid the PPA trails KEPCO even without termination and falls further on a break — it must last 8+ years to hold 8%

Equity IRR by termination year → hurdle-holding term → Vol.2 credit → contractual mitigants

Equity IRR by termination year: at the corridor mid the PPA trails KEPCO (12.2%) even if it runs to term *1

Unit: Equity IRR



Source: *1 KSJ Research
Note: x-axis = termination year N (SMP 113.2 from year N = PPA lasted N−1 years). PPA at corridor mid KRW 149.5/kWh, debt fixed (no refinancing), no termination payment

To keep the 8% hurdle the PPA must last 8+ years *1

Termination	Equity IRR	vs KEPCO
Year 1 (PPA lasted 0 yrs)	3.1%	below
Year 3 (PPA lasted 2 yrs)	4.2%	below
Year 5 (PPA lasted 4 yrs)	5.5%	below
Year 8 (PPA lasted 7 yrs)	7.5%	below
Year 10 (PPA lasted 9 yrs)	8.6%	below
Year 15 (PPA lasted 14 yrs)	10.5%	below
Year 20 (PPA lasted 19 yrs)	11.4%	below
No termination	11.6%	—

Source: *1 KSJ Research
Note: After termination the plant earns SMP, so early breaks hurt most — termination payments prop up the left side of this curve

Offtaker = Vol.2's data centre: lender min DSCR 1.63x, power passed to tenants *1

Item	Value
Data-centre SPV min DSCR (Vol.2 Korea)	1.63x
Data-centre equity IRR (Vol.2)	16.1%
Power cost treatment (Vol.2)	Billed to tenants (100%)
PPA volume / data-centre use	15%

Source: *1 KSJ Research Vol.2 (Oct 2026)
Note: The PPA is ultimately paid from tenants' power bills — if leases are shorter than the PPA, the gap becomes termination risk

Mitigants: contract terms that cut termination risk and loss *1

What to hold in the term sheet (A1)

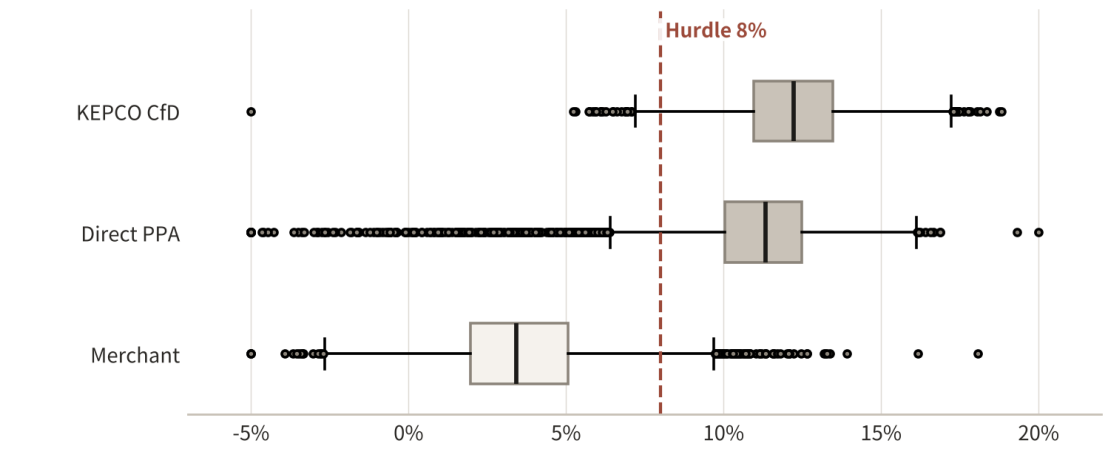
- Termination payment: covers part of the PPA-vs-SMP gap for the remaining term
- Credit support: parent guarantee or LC, or align lease power clauses with the PPA term
- Assignment: the PPA follows the data centre on sale or refinancing
- Change in law: a renegotiation rule if network-charge reform changes the buyer's cost

Source: *1 KSJ Research

Monte Carlo: PPA equity IRR P50 11.3% · P5 5.5% — below KEPCO 99% of the time; merchant loses money 5% of the time

5,000 runs × 3 routes · seed 42 · SMP (log AR1) × output × PPA termination

Equity IRR (5,000 runs): the PPA has a lower median and a weaker bottom 5% (P50 11.3% vs 12.2% · P5 5.5% vs 9.1%) *1,*2



Source: *1 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *2 KSJ Research
Note: Common random numbers (SMP, output) across routes — differences come from the contracts only. PPA termination 1% a year. Shown −5% to 20%

Probability below the 8% hurdle: KEPCO 1% · PPA 9% *1

Route	P5	P10	P50	P90	< hurdle	Loss (IRR<0)
KEPCO CfD	9.1%	9.9%	12.2%	14.7%	1%	0%
Direct PPA	5.5%	8.3%	**11.3%**	13.5%	9%	1%
Merchant	0.0%	0.7%	3.4%	6.7%	96%	5%

Source: *1 KSJ Research
Note: The PPA underperforms KEPCO on the same path 99% of the time. Debt fixed at base sizing — equity funds shortfalls (no lock-up or default mechanics in the model)

Inputs: SMP volatility 0.23 and mean reversion 0.72 estimated from 24 years of annual SMP (2002–2025) *1

Input	Value	Grade
SMP volatility	0.23	Computed from official data
SMP mean reversion (AR1)	0.72	Computed from official data
Annual output variation	5%	Author judgement
Resource estimate error	4%	Author judgement
PPA annual termination	1%	Author judgement
Runs · seed	5,000 · 42	—

Source: *1 KSJ Research
Note: SMP: log AR(1), mean-preserving · output: one resource error × annual noise · termination: geometric within the PPA term

How to read it: the PPA's risk is in the tail, not the average *1

What the distribution says

- KEPCO: narrow (P10 9.9%–P90 14.7%) — only output varies
- PPA: at the corridor mid the distribution sits below KEPCO (P50 11.3%, P5 5.5%) — worse than KEPCO on 99% of paths
- Merchant: 5% loss probability — uncontracted new solar is not an equity case
- IC view: settle the price (the buyer's RE100 willingness to pay) first, then tail protection (termination payments, guarantees)

Source: *1 KSJ Research

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Implications: measured on solar-hour rates the price itself blocks the deal — confirm the buyer's RE100 willingness to pay first, then charges and termination protection

Conclusion · IC checklist · limits · Vol.5 preview

Conclusion: on solar-hour rates the direct-PPA price does not close [*1,*2,*3]

- Price: the seller needs +5.1–16.5 over KEPCO (1% termination, 0–2pp spread) — the buyer saves only +3.6 in solar hours
- Make-or-break: buyer room −12.9 ~ −1.4 on the solar-hour saving — covering ~40 of charges needs 41–53 per kWh of RE100 value from the buyer
- Risk: early termination — termination payments, guarantees and assignment lower the loss and the premium
- Alternatives: KEPCO at 12.2% (above the corridor-mid PPA's 11.6%), merchant at 3.7%

Source: *1 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *2 SFOC, 'Stranded RE100' (Dec 2021, citing KEPCO's third-party PPA charge simulator) *3 KSJ Research

IC checklist: this note's assumption vs what a real deal must confirm [*1]

Item	This note	Confirm in a deal	p.
Price	Corridor 147.7–151.3	2027 contract-market cap · actual tariff class and option · measured hourly output · climate and fuel add-ons	3, 5
Network charges	Room −13 ~ −1 vs 40 example	Current T&D tariffs · settlements · losses · fund levy · buyer's RE100 willingness to pay	3, 5
Offtaker	Termination grid 0–5%	Rating · lease term · guarantees	6
Output	CF 16% (BNEF/KEEI)	P50/P90 yield report · curtailment	4, 7
CAPEX	KRW 61.1bn (from LCOE)	EPC quote · land · grid connection	4
Shape	Pay-as-produced	24/7 firming cost (storage, KEPCO)	—

Source: *1 KSJ Research

Limits of this note [*1]

- CAPEX: 2022 actual installed cost (KRW 1,396k/kW) implies an LCOE of 100, below KEEI's 115–136 — matched to the LCOE to stay conservative
- Buyer ceiling = solar-hour energy rate (HV-A option II): daily shape and Saturday/holiday rules are judgements, climate and fuel add-ons excluded, weekend discount ends 2030 (154.1 after) · no RE100 value or tariff rises
- Plant sized at 35 MW so all output is used — 24/7 shape risk, storage and curtailment left out
- Debt terms, termination rate and risk spread are unsourced judgements — shown as grids and sensitivities

Source: *1 KSJ Research

Next, Vol.5 (finale): an AI energy asset investment map [*1]

Demand, equipment, PF, IPO and power on one page

- Place Vol.1–4 on one risk-return map — which asset fits which investor
- IC memo format: thesis, risks and mitigants, term sheet
- All assumptions and source grades in one table

Source: *1 KSJ Research

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IC memo: equity in a direct-PPA plant — on hold; KEPCO is the base case until the buyer is shown to pay 11+ per kWh for RE100 value (KPX settlement only)

Summary · points · risks and mitigants · term sheet (illustrative)

Deal summary (illustrative — hypothetical asset) *1

Item	Detail
Asset	Hypothetical 35 MW solar, COD 2028
Revenue	20-yr direct PPA with a data centre, pay-as-produced
Uses · sources	KRW 61.1bn = debt KRW 45.8bn (75%) + equity KRW 15.3bn
Equity return	IRR 11.6% (corridor mid) · Monte Carlo P50 11.3%
Pricing rule	At least KRW 152.8/kWh–KRW 164.2/kWh (KEPCO +5–16, 1% termination, 0–2pp spread)

Source: *1 KSJ Research

Three investment points *1

- No price edge: at the corridor mid, −0.7pp IRR vs KEPCO — matching KEPCO needs KRW 152.3/kWh, above the buyer's KRW 151.3/kWh ceiling
- Demand: AI data centres need RE100 supply — direct routes get scarcer after the 2027 RPS repeal
- Downside: terminated in year 1 and selling at SMP 113, min DSCR 1.06x (debt paid, lock-up territory); at SMP 90 it falls to 0.80x — termination payments and guarantees are essential

Source: *1 KSJ Research

Risks and mitigants *1

Risk	Mitigant
Offtaker termination	Termination payment · parent guarantee/LC · assignment
Charges exceed the buyer's room	Fix the buyer's all-in price with current T&D tariffs and settlements before signing; change-in-law reopener
Output shortfall	P90 debt sizing · curtailment compensation
SMP fall (after termination)	Re-contracting rights · re-bidding into the KEPCO market
Policy change	Check transition rules (pre-2027 commissioning)

Source: *1 KSJ Research

PPA term sheet (illustrative) *1

Clause	Proposal
Term	20 yrs (aligned with leases)
Price	At least KRW 152.8/kWh fixed (1% break-even 152.8; corridor mid 149.5 falls short)
Volume	All output (pay-as-produced)
Termination	A share of the PV of (PPA − SMP) over the remaining term
Credit support	Parent guarantee or a one-year LC
Attributes	Renewable certificates to the buyer — their value must bridge the gap to the buyer's saving

Source: *1 KSJ Research
Note: Illustrative terms, not an actual contract

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Inputs and source grades (1/2): every number carries a grade

Value · range · unit · grade · source (links in the repository input file)

56 inputs: official 16 · computed from official 3 · press report 4 · secondary 0 · judgement 33 *1,*2,*3,*4,*5,*6

Item	Value (range)	Unit	Grade	Source / basis
Capacity	35	MW	Author judgement	저자 설계값: 가상 태양광 35MW(AC). 최대 출력이 Vol.2 한국 데이터센터 평균 부하(IT 40MW × 70% × PUE 1.3 = 36.4MW)를 넘지…
COD year	2028	년	Author judgement	저자 설계값: 2026년 개발 착수 → 2028년 상업운전(신규 사업자는 2027-01-01부터 RPS 대신 계약시장 적용)
Installed cost per kW (2022 actual)	1,396 (1,396~1,635)	천원/kW	Official source	에너지경제연구원 기본연구 25-04 「재생에너지 보급비용 최소화를 위한 태양광 설치비용 영향요인 실증 연구」(공지영·장연재) PDF 93쪽: 국내 태양광 실제 설…
Capacity factor	0.16 (0.14~0.18)	비율	Official source	에너지경제연구원 기본연구 25-04 「재생에너지 보급비용 최소화를 위한 태양광 설치비용 영향요인 실증 연구」(공지영·장연재) PDF 10쪽 요약·31쪽 <표 2-…
Annual degradation	0.005 (0.003~0.008)	연 비율	Official source	NREL(미 국립재생에너지연구소) Jordan & Kurtz, Photovoltaic Degradation Rates — An Analytical Review(2…
Solar LCOE	125.5 (115~136)	원/kWh	Official source	에너지경제연구원 기본연구 25-04 「재생에너지 보급비용 최소화를 위한 태양광 설치비용 영향요인 실증 연구」(공지영·장연재) PDF 9·18쪽: 2024년 국내 …
WACC (LCOE cross-check)	0.06	연 비율	Official source	에너지경제연구원 기본연구 25-04 「재생에너지 보급비용 최소화를 위한 태양광 설치비용 영향요인 실증 연구」(공지영·장연재) PDF 10쪽·31쪽 <표 2-1>:…
OPEX (% of CAPEX)	0.015 (0.01~0.02)	CAPEX 대비 연 비율	Author judgement	저자 판단: 운영·유지비 + 보험(CAPEX 대비 연 1.5%). 에너지경제연구원 설치비 연구 원문에 운영비 수치 없음 — 공식 근거 미확보로 민감도(1~2%) 공개
OPEX escalation	0.02	연 비율	Author judgement	저자 판단: 물가 수준
Operating life	25	년	Author judgement	저자 판단: 태양광 모듈 보증 25년 수준
Depreciation life	20	년	Author judgement	저자 판단: 정액 20년 (법인세법 시행규칙 기준내용연수 확인 필요)
Corporate tax brackets	2:10,200:20,3000:22,inf:25	억원:%	Official source	법인세법 제55조(2026 사업연도부터: 과세표준 2억원 이하 10%, 200억원 이하 20%, 3,000억원 이하 22%, 초과 25%) — Vol.2와 같음,…
Local income tax	0.1	법인세 대비	Official source	지방세법 — 법인지방소득세(법인세의 10%)
KEPCO contract price (2026 round-1 cap)	147.7	원/kWh	Author judgement	저자 판단(대리값): 기후에너지환경부 보도자료 '2026년도 1차 태양광 경쟁입찰 공고'(2026-07-16) 원문의 입찰 상한가격 147.686원/kWh(공식 …
Contract term	20	년	Press report / secondary report	기후부 계약시장 개편(2026-09-15 공포, 2027-01-01 시행): 한전과 장기 고정가격 계약 — 파이낸셜뉴스 보도. 20년은 고정가격계약 관행
KEPCO industrial average tariff (2025, reference)	181.9	원/kWh	Official source	한전 전력통계월보 제566호(2025.12) 표 18·18-2: 2025.01~12 산업용 판매수입 50,971,210백만원 ÷ 판매전력량 280,221,380M…
Industrial (B) time-of-use rates	산업용(을) 개편 후	원/kWh	Official source	기후에너지환경부·한전 보도참고자료 「계절·시간대별 전기요금 체계 개편안 공개」(2026-03-13) PDF 8쪽(전체 9쪽) [붙임2]: 개편 후 시간대 구분(봄…
Buyer's tariff class	고압A 선택 II	계약종별	Author judgement	저자 판단: Vol.2와 같은 산업용 가정(데이터센터 평균 부하 약 36MW, 300kW 이상 산업용(을)). 고압A 선택 II 전력량요금 = 봄가을 경부하 116.…
Spring/autumn weekend midday discount	0.5	할인율	Official source	같은 보도참고자료 PDF 3쪽·7쪽 [붙임1]: 봄(3~5월)·가을(9~10월) 토·일요일·공휴일 11~14시 전력량요금 50% 할인, 2030-12-31까지(연…
Sunday/holiday rate rule	일요일 전 시간 경부하	규칙	Author judgement	PDF 2쪽 그림(고압A 선택2·봄가을 예시): 일요일·공휴일은 하루 종일 경부하 요금(11~14시는 50%). 같은 규칙을 여름·겨울에도 적용(저자 판단). 평…
Saturday time-band rule	평일과 같음	규칙	Author judgement	저자 판단: 개편 보도자료에 토요일 시간대 규칙이 없어 평일 시간대를 그대로 적용(봄·가을 11~14시 50% 할인만 반영). 민감도: 토요일 최대부하를 중간부하…
Monthly solar output weights	2025	년	Computed from official data	한전 전력통계월보 제566호 시트 '16-1.태양광 발전량 추계정보': 2025년 월별 태양광 발전량(전력시장 참여 + 한전PPA + 자가용 추계, 합계 45,4…
Daily output shape exponent	1 (0.5~2)	지수	Author judgement	저자 판단(표준 형상): 저장소에 공식 시간별 태양광 발전량이 없어 하루 발전 형상 = max(cos 천정각, 0)^p. 천정각은 Spencer(1971) 적위·…
Per-kWh add-ons (climate, fuel adjustment)	미포함	원/kWh	Author judgement	기후환경요금·연료비조정요금 등 kWh당 추가 요금: 저장소에 공식 원문이 없어 회피 요금에 넣지 않음(넣으면 구매자 상한이 그만큼 오르는지는 직접 PPA 정산 규…
Ancillary charges, public example (3rd-party PPA, 2021)	40	원/kWh	Press report / secondary report	기후솔루션(SFOC) 「좌초되는 RE100 제도」(2021-12) 표: 한전 에너지마켓플레이스 '제3자간 계약 요금계산 시뮬레이션'으로 본 중대형 태양광(고압 배…
Direct-PPA ancillary settlement (Jul 2023–Jun 2024)	9.73	원/kWh	Official source	한국전력거래소 홈페이지 질의응답(2023-12-19): 직접전력거래(직접 PPA) 부가정산금 단가 9.73원/kWh(2023.7~2024.6 적용) — 부대비용 …
SMP base	113.2 (90.4~148.4)	원/kWh	Official source	전력거래소 EPSIS 계통한계가격(가중평균 SMP, 월별 육지) 엑셀: 2025-09~2026-08 육지 SMP 12개월 평균 113.2원/kWh(범위 = 같은 …
SMP trend	0	연 비율	Author judgement	저자 판단: 명목 SMP 추세 0% (민감도로 공개)

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *4 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *5 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *6 KSJ Research
Note: Grades: official (gold) = government, public body, statute or SEC original · press report = media citing an official release · secondary (red) = replace before final · judgement (brown) = stated basis, stress-tested

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Inputs and source grades (2/2): every number carries a grade

Value · range · unit · grade · source (links in the repository input file)

56 inputs: official 16 · computed from official 3 · press report 4 · secondary 0 · judgement 33 *1,*2,*3,*4,*5,*6

Item	Value (range)	Unit	Grade	Source / basis
SMP volatility	0.229	로그 표준편차(연)	Computed from official data	전력거래소 EPSIS 계통한계가격(가중평균 SMP, 월별 육지) 엑셀: 2002~2025 연평균 SMP(24년) 로그 AR(1) 회귀 잔차 표준편차 0.229 (…
SMP mean reversion (AR1)	0.718	AR(1) 계수	Computed from official data	전력거래소 EPSIS 계통한계가격(가중평균 SMP, 월별 육지) 엑셀: 같은 회귀의 AR(1) 계수 0.718 — 연료비 사이클로 평균 회귀
REC revenue	0	여부	Press report / secondary report	신규 사업자는 2027-01-01부터 REC 발급 제한, REC 현물시장은 2029년 말까지 — 기후에너지환경부 공청회(2026-09-30) 내용 보도(다음 뉴스…
Annual output variation	0.05	비율	**Author judgement**	저자 판단: 연간 일사량 변동(해마다 독립)
Resource estimate error	0.04	비율	**Author judgement**	저자 판단: 장기 발전량 추정 오차(P50 오차)
KTB 3-yr	3.84	%	**Official source**	한국은행 ECOS 국고채 3년 (기준일 값)
Credit margin	150	bp	**Author judgement**	저자 판단: Vol.2 한국 케이스와 같은 가산금리
Target DSCR · CfD	1.2	배	**Author judgement**	저자 판단: 정부 계약(한전 상대방) 매출 — 계약 매출 PF의 낮은 쪽
Target DSCR · PPA	1.3	배	**Author judgement**	저자 판단: 민간 오프테이커(데이터센터 SPV) 신용 위험만큼 높임
Target DSCR · merchant	1.6	배	**Author judgement**	저자 판단: 시장가 매출 — 가격 위험 반영
Max gearing · CfD	0.8	비율	**Author judgement**	저자 판단
Max gearing · PPA	0.75	비율	**Author judgement**	저자 판단
Max gearing · merchant	0.55	비율	**Author judgement**	저자 판단
Tenor · CfD	18	년	**Author judgement**	저자 판단: 계약 20년 안에서 2년 꼬리(tail)
Tenor · PPA	18	년	**Author judgement**	저자 판단: PPA 20년 안에서 2년 꼬리
Tenor · merchant	12	년	**Author judgement**	저자 판단
PPA term	20	년	**Author judgement**	저자 판단: 20년 — 해외 하이퍼스케일러 원전 PPA 20년 사례(Meta–Constellation Clinton, Utility Dive 2025-06; Am…
PPA escalation	0	연 비율	**Author judgement**	저자 판단: 고정가(민감도로 공개)
Equity hurdle	0.08	연 비율	**Author judgement**	저자 판단: 지분 요구수익률(엑싯 가치 할인·손실 확률 기준)
Exit year	10	년	**Author judgement**	저자 판단: 인프라 펀드 보유기간
Buyer's required return	0.07	연 비율	**Author judgement**	저자 판단: 운영 단계 자산 매수자 요구 지분수익률(엑싯 가격 산정)
Data-centre IT capacity	40	MW	**Author judgement**	KSJ Research Vol.2 한국 케이스 설계값(1단계 20MW + 2단계 옵션 20MW, 저자 설계)
PUE	1.3	배	**Author judgement**	KSJ Research Vol.2 가정(저자 판단)
IT load factor	0.7	비율	**Author judgement**	KSJ Research Vol.2 가정(저자 판단)
Talen–AWS PPA notional revenue	18	USD bn	**Official source**	Talen Energy 8-K 투자자 자료(2025-06-11) 3·7쪽: AWS PPA 17년 계약, 전량 기준 명목 매출 약 180억 달러(~$18bn not…
Talen–AWS term	17	년	**Official source**	Talen Energy 8-K 투자자 자료(2025-06-11) 3쪽: 17년 계약
Talen–AWS contract capacity	1,920	MW	**Official source**	Talen Energy 8-K 보도자료(2025-06-11): 전량 1,920MW, 2042년까지, 2032년까지 단계적 증가
LevelTen N. America solar PPA	61.4	USD/MWh	Press report / secondary report	LevelTen Energy 2026년 2분기 북미 태양광 PPA 시장 평균 $61.40/MWh — TaiyangNews 보도(2026-07-23). 원문 지수 …

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *4 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *5 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *6 KSJ Research
Note: Grades: official (gold) = government, public body, statute or SEC original · press report = media citing an official release · secondary (red) = replace before final · judgement (brown) = stated basis, stress-tested

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Data notes: official sources first; everything else flagged for replacement

As of 2026-10-20 · KTB 3-yr 3.84% (2026-10-20) · annual nominal model, KRW 100mn, KRW/kWh

7 inputs awaiting official sources — all replaced before final [*1]

Item	Current grade	Official source to use
Contract term	Press report / secondary report	기후에너지환경부 보도자료 원문(2026-09-15) 계약기간 확인
Sunday/holiday rate rule	Author judgement	한전 기본공급약관 시행세칙(일요일·공휴일 시간대 적용 기준) · 행정안전부 관공서 공휴일 목록(2025)
Saturday time-band rule	Author judgement	한전 기본공급약관 시행세칙 — 토요일 최대부하 시간대 적용 기준
Daily output shape exponent	Author judgement	공공데이터포털 '한국전력거래소 시간별 태양광 발전량' — 표준 형상을 실측 형상으로 교체
Per-kWh add-ons (climate, fuel adjustment)	Author judgement	한전 전기요금표 원문(기후환경요금·연료비조정요금 단가) · 직접 PPA 구매자의 부과 여부
REC revenue	Press report / secondary report	기후에너지환경부 보도자료 원문
LevelTen N. America solar PPA	Press report / secondary report	LevelTen 원문 보도자료

Built with an AI coding tool (Claude Code) for cash flows, debt sizing, break-evens, Monte Carlo and report generation; question design, source grading, assumption judgments and number checks by the author.

Source: *1 KSJ Research
Note: Rule: official originals first (government, public bodies, statutes, SEC) · media only when citing an official release · private data provisional only. Code and links: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Buyer-ceiling sensitivity: saving 148.2–162.0 — the tariff class moves it more than the daily shape [*1,*2,*3]

Assumption	Annual saving (KRW/kWh)	Room (vs required 152.8)
Base: HV-A option II · shape p = 1 · Saturday as weekday	**151.3**	−1.4
Daily shape p = 0.5	152.0	−0.8
Daily shape p = 2	149.8	−2.9
Class: HV-A option I	156.7	+3.9
Class: HV-A option III	148.2	−4.6
Class: HV-B option II	162.0	+9.3
Saturday peak → mid-peak	149.8	−3.0
Spring/autumn weekend discount ends (2031+)	154.1	+1.3
Ref.: pre-reform rates (to 15 Apr 2026)	167.0	+14.2

Source: *1 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *2 KEPCO Monthly Statistics no.566, table 16-1 (2025 monthly solar output) *3 KSJ Research
Note: Required PPA at 1% termination, no spread. One assumption at a time. Climate and fuel add-ons excluded throughout (no official source in hand)

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Glossary: the language of buying and selling power

Prices KRW/kWh · amounts KRW 100mn · U.S. USD/MWh

Glossary

Term	Definition
PPA	Power purchase agreement — seller and buyer fix price, term and volume directly
CfD	Contract for difference — settles contract vs market price; from 2027, long-term contracts with KEPCO
SMP	System marginal price — Korea's hourly wholesale power price
REC	Renewable energy certificate under RPS; restricted for new entrants from 2027
RPS	Renewable portfolio standard — replaced by a contract market in 2027
LCOE	Levelised cost of electricity — PV of lifetime costs ÷ PV of output
Network charges	Transmission and distribution charges a direct-PPA buyer pays KEPCO
Offtaker	The buyer under the contract (here the data-centre SPV)
DSCR	Debt service coverage = CFADS ÷ debt service
Sculpting	Shaping debt service so each year's DSCR equals the target
Equity IRR · MOIC	Return on equity · multiple of invested capital
Termination rate	Annual chance the PPA ends early (shown as a grid)
Price corridor	Range between the seller's (KEPCO contract) and buyer's (solar-hour KEPCO energy rate) outside options
Avoided energy rate	The KEPCO rate a PPA kWh replaces — time-of-use energy rates in solar hours, output-weighted (no demand charge)
Pay-as-produced	Sell what is produced — no 24/7 obligation
Source grade	Official · computed from official data · press report of official release · secondary · judgement

Note: Shaded = core terms

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- The 35 MW solar asset and the data-centre offtaker are hypothetical; no actual contract prices or terms are shown.
- Views are the author's own and do not represent any current or future employer.
- Every input carries a source grade (official · computed from official data · press report of official release · secondary · judgement); see Appendix A2–A4.
- As of 2026-10-20. Methods and code can be reproduced from the repository.
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https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

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